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MIL-I-2781D
INTERIM AMENDMENT-3 (SHIPS)
1 March 1971 1/
SUPERSEDING
INTERIM AMENDMENT-2 (SHIPS)
16 July 1965

MILITARY SPECIFICATION

INSULATION, PIPE, THERMAL

This Interim Amendment is issued for use by the Naval Ship Engineering Center with Military Specification MIL-I-2781D dated 13 June 1963.

Page 1

1.2: Under Grade II, delete "Class c - Fibrous"; and Grade III, delete "Class f - Fibrous."

2.2: Delete all reference to "OFFICIAL CLASSIFICATION COMMITTEE" and substitute the following:

"UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)"

Page 2

3.1, Delete and substitute:

"3.1 Qualification. The thermal pipe insulation furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.2)."

Page 3

4.1, Delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

4.2, line 2: Delete "Bureau of Ships" and substitute "Naval Ship Engineering Center".

Footnote 1: After "Provisions Governing Qualification" add "SD-6".

1/ CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 5640

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Page 5

Table III, delete and substitute:

"Table III - Physical requirements.

Characteristic	Test paragraph	Grade I Class b	Grade II Class d	Grade III Class e Type I	Grade III Class e Type II
Density average, lbs/cu.ft. maximum	4.4.2	14.0	14.0	14.0	22.0
Thermal conductivity (average) B.t.u. in. hr sq ft °Fahr. at mean temperature of:	4.4.3				
200°F.		.40	.40		
300°F.		.45	.45		
400°F.		.50	.50		
500°F.		---	---	.60	.66
600°F.		---	---	.65	.71
700°F.		---	---	.70	.76
Weight loss by tumbling	4.4.4				
Loss in weight (average), percent max.					
After first 10 minutes		40	40	40	55
After second 10 minutes		70	70	70	80
Modulus of rupture (average), lbs./sq.in., minimum	4.4.5	1/	1/	1/	2/
Changes under soaking heat for 6 hours at °F.	---	600	750	1,200	1,200
Loss in weight (average), percent max.	4.4.6	18.0	12.0	12.0	16.0
Linear shrinkage (average), percent, max.	4.4.6	2.0	2.0	2.0	2.0

1/ Three times density (lbs. per cu. ft.) of the sample tested.
2/ Two and one-half times density (lbs. per cu. ft.) of the sample tested."

4.3.1.1.1, line 3: Delete "I" and substitute "II", and delete "4.0 percent" and substitute "2.5 percent".

4.3.1.1.2, line 3: Delete "L-6" and substitute "II".

Page 6

Table IV, delete and substitute:

"Table IV - Sampling for tests.

Lot size in half section or segments	Sample size = Number of test specimens for each test (4.4.2, 4.4.4, 4.4.5, 4.4.6)	Number of test failures allowed (each test)
Up to 63	None	-
64 to 160	2	0
161 to 400	3	0
401 to 1,000	5	0
1,001 to 2,500	8	0
2,501 to 6,300	13	1
6,301 to 16,000	20	2
16,001 to 40,000	32	3

"

Page 8

6.2 and 6.3: Delete and substitute:

"6.2 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 2781 whether or not such products have actually been listed by that date. The attention of the suppliers is called to this requirement, and

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manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3).

"6.3 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120."

Add new paragraph 6.4 as follows:

"6.4 Class c and class f have been eliminated from this specification because of their high asbestos content. The remaining classes are considered equivalent in performance."

Preparing activity:
Navy - SH
(Project 5640-N013)

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